

Factor Model of La Liga Player Performance: Technical Report

Hierarchical Bayesian Analysis — K=3 Student- t Model with Minutes-scaled
Residuals

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i Note**What changed since the last version (August 2026)**

Five additions since the previous report:

1. **Playing-time heteroskedasticity fixed** — diagnosed that 48.9% of worst residuals came from low-minute players; extended the likelihood to scale residual variance by $\sqrt{\bar{m}/m_n}$; confirmed the fix drops the low-minute share to 8.3% (§2).
2. **Factor loadings now have posterior CIs** — draws-level PCA of $\Lambda\Lambda^\top$ gives 90% credible intervals on each loading, replacing earlier point-estimate plots (§3.2).
3. **Rank $K=3$ confirmed by player-holdout CV** — 15% player holdout cross-validation ($K=1-6$) peaks at $K=3$ under both constant and minutes-scaled residual metrics (§3).
4. **Player profile radars + similarity metric** — FIFA/FC-style percentile radars for individual players and pairs; Euclidean archetype-space similarity with named examples (§3.4).
5. **Season profiles updated** — group-trend ribbons, heatmap, and new top-5 most-changed-features figure (§3.5).

1 The Model

1.1 Specification

We observe $N = 4,586$ player-season records: $I = 1,529$ outfield players, $S = 12$ La Liga seasons (2011–2023), $P = 48$ Z-scaled performance features (goalkeepers excluded; 23 features transformed with log1p or sqrt before scaling to reduce right skew).

The additive decomposition is:

$$y_n = a_{i[n]} + b_{j[n]} + \varepsilon_n$$

- $a_i \in \mathbb{R}^P$: **player effect** — stable latent identity
- $b_j \in \mathbb{R}^P$: **season effect** — league-wide shift in season j
- ε_n : **residual** — observation noise

1.2 Covariance and likelihood

Player effects share a low-rank plus diagonal covariance:

$$\Sigma_a = \Lambda\Lambda^\top + \Psi_a, \quad \Lambda \in \mathbb{R}^{P \times K}, \quad K = 3$$

Season and residual covariances are diagonal. Residuals follow Student- t with fitted degrees of freedom and, in the current production model, playing-time-scaled scale:

$$\varepsilon_{np} \sim t\left(\nu, 0, \sigma_{e,p} \cdot \sqrt{\frac{\bar{m}}{m_n}}\right), \quad \nu \sim \text{Gamma}(2, 0.1)$$

where m_n is player-season n 's minutes and $\bar{m} = 1,585$ min is the sample median.

1.3 Identification and priors

Λ is constrained lower-triangular with positive diagonal (LT-PD), removing rotational non-identifiability. Factor directions are interpreted post-hoc via PCA of $\hat{\Lambda}\hat{\Lambda}^\top$ (§3.2). Player effects are parametrised non-centredly ($a_i = \Lambda\eta_i + \psi_a \odot z_{u,i}$, raw scores centred across players). Priors: $\Lambda_{kk} \sim \text{HN}(1)$; $\Lambda_{pk} \sim \mathcal{N}(0, 1)$ for $p > k$; $\psi_{a,p}, \sigma_{b,p}, \sigma_{e,p} \sim \text{HN}(1)$. NUTS via CmdStan 2.35: 4 chains, 1,000 warmup + 1,000 post-warmup draws, $\delta = 0.95$, PCA-based initialisation.

2 Playing-time Heteroskedasticity

2.1 What we found

Performance features are per-90 rates. A player who plays 500 minutes in a season contributes estimates based on ≈ 6 matches; one who plays 3,000 minutes contributes 35 matches. The ratio-estimator variance of a per-90 rate scales as $1/m_n$, so low-minute observations are inherently noisier — yet the prior model used a constant $\sigma_{e,p}$ for all player-seasons.

A quantile diagnostic on the K=3 t-model (constant σ_e) confirmed this: **48.9% of the top-200 worst standardised residuals came from the bottom 20th percentile of playing minutes** (expected: 20%; binomial $p < 10^{-17}$).

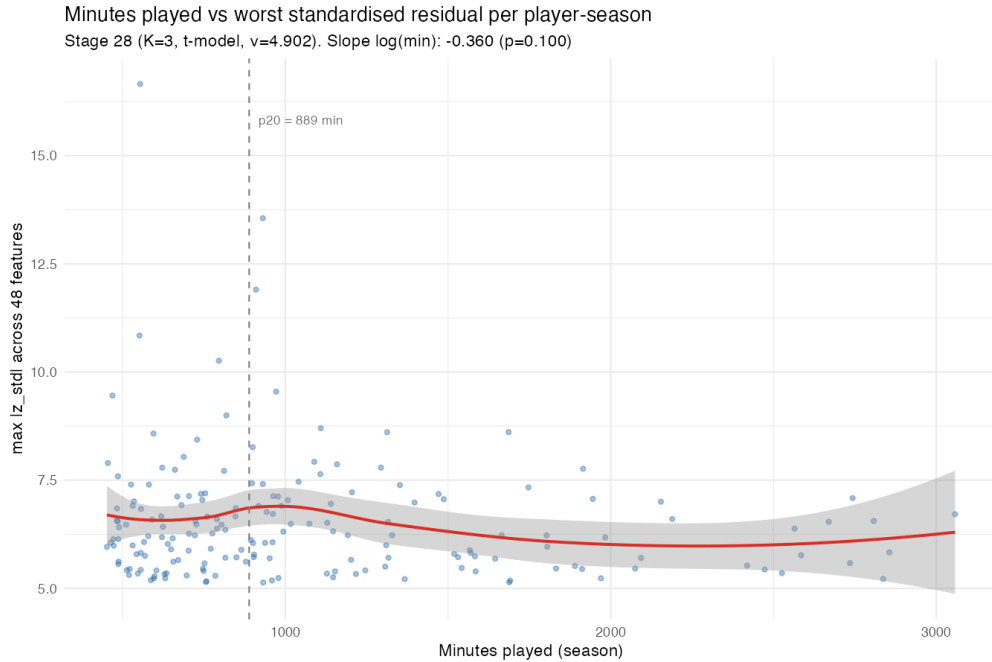


Figure 1: Worst-residual cells vs playing minutes (K=3 t-model, constant σ_e). Points in red are in the top-200 worst cells by $|z_{np}|$; points in grey are all other observations. The concentration of extremes at low minutes (left of dashed 20th-percentile line) confirms the heteroskedasticity problem.

2.2 The fix: minutes-scaled residual variance

For Poisson-rate data, $\text{Var}(X/m) = \lambda/m$, so residual SD $\propto m^{-0.5}$ ($\phi = 0.5$). We modify the likelihood to:

$$\varepsilon_{np} \sim t\left(\nu, 0, \sigma_{e,p} \cdot \sqrt{\frac{\bar{m}}{m_n}}\right)$$

The reference scale $\sigma_{e,p}$ is at median playing time, so ICC decomposition is unchanged at $m = \bar{m}$. A second variant estimates ϕ directly with prior $\phi \sim \mathcal{N}(0.5, 0.2)$, testing whether the Poisson prediction $\phi = 0.5$ holds.

2.3 Results

	Constant σ_e (prev. model)	Fixed $\phi = 0.5$	Estimated ϕ
$\hat{\nu}$	4.902	5.667	5.979
95% CI for ν	[4.79, 5.01]	[5.54, 5.80]	[5.83, 6.13]
$\hat{\phi}$	—	0.5 (fixed)	0.366
95% CI for ϕ	—	—	[0.360, 0.373]
ESS _{bulk} (ν)	2,734	2,949	2,415
$\hat{R}$ (ν)	1.000	1.000	1.001

The rise in $\hat{\nu}$ (from 4.90 to 5.67–5.98) confirms the diagnosis: once low-minute variance is explicitly modelled, the Student- t tail absorbs less unexplained noise. Student- t remains warranted — $\hat{\nu} \neq \infty$.

$\hat{\phi} = 0.366 < 0.5$: the estimated exponent is below the Poisson prediction. This is consistent with *super-Poisson* variance — player heterogeneity is partially absorbed by the player factor a_i , which reduces the residual variance exponent. Fixed $\phi = 0.5$ (the Poisson-theory default) is the production model; the estimated variant confirms the direction and that scaling is real.

2.4 Validation

We re-run the same diagnostic on the minutes-scaled model’s residuals:

$$z_{np}^{35} = \frac{y_{np} - \hat{a}_{i[n],p} - \hat{b}_{j[n],p}}{\hat{\sigma}_{e,p} \sqrt{\bar{m}/m_n} \sqrt{\hat{\nu}/(\hat{\nu} - 2)}}$$

Diagnostic	Before (constant σ_e)	After (minutes-scaled, $\phi = 0.5$)
Low-minute share in top-200 worst	48.9% (88/180 obs)	8.3% (14/168 obs)
Binomial p (H1: share > 20%)	$< 10^{-17}$	> 0.999
Regression slope: $\max, z \sim$	negative, $p \ll 0.001$	0.007, $p = 0.97$
log,min		
Verdict	Over-represented	CORRECTED

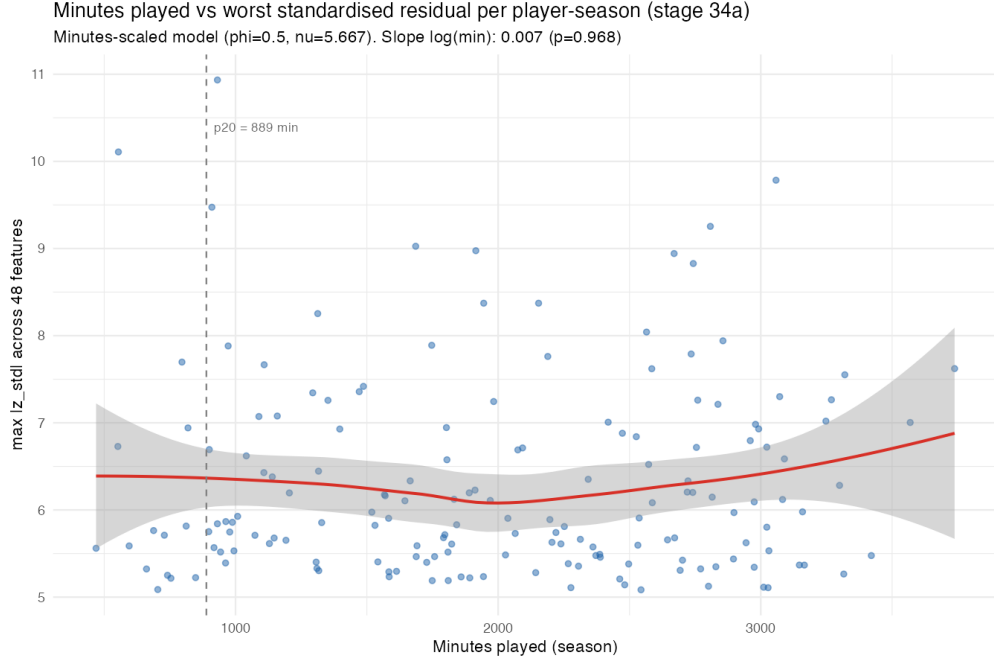
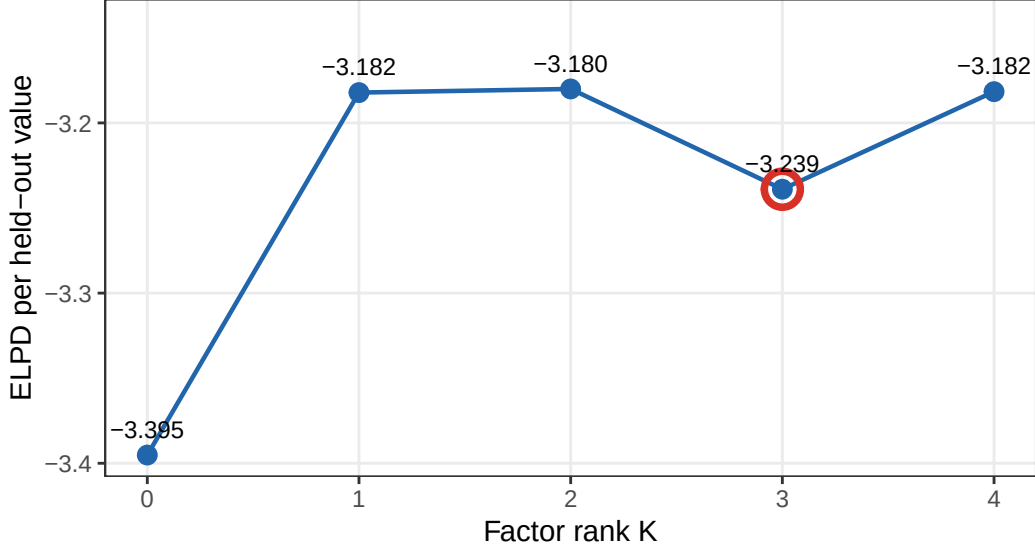


Figure 2: Worst-residual cells vs playing minutes after minutes-scaling ($\phi = 0.5$). Red points are top-200 worst cells under the minutes-scaled model. The low-minute concentration is eliminated — the red points are spread uniformly across the minutes range.

3 Factor Analysis Results

3.1 Rank selection: $K=3$ confirmed by two methods

Pathfinder screening ($K = 0, \dots, 4$): fast variational approximation on the full dataset. ELPD rises $K=0 \rightarrow 3$ then reverses at $K=4$.



Circle = selected $K^*=3$. Pathfinder approximate; direction reliable.

Figure 3: Pathfinder ELPD per held-out value for $K = 0, \dots, 4$ (rebuilt data, $N = 4,586$, $P = 48$). Reversal at $K=4$ selects $K^* = 3$.

Player-holdout CV ($K = 1, \dots, 6$): 15% of players (228 players, stratified) are withheld entirely. A Normal marginalised model (integrating out a_i analytically via Woodbury identity) is fitted with Pathfinder for each K . Holdout ELPD evaluated for two residual metrics: constant σ_e and minutes-scaled $\sigma_e \sqrt{\bar{m}/m_n}$.

Table 3: Player-holdout ELPD by factor rank. Both metrics peak at $K^* = 3$, confirming the Pathfinder selection. $K=7-8$ not computed (wall-clock constraint).

K	ELPD (base)	ELPD (minutes-scaled, $\phi = 0.5$)
1	-34755.3	-34220.5
2	-34758.6	-34216.0
3	-34737.2	-34211.0
4	-34763.4	-34209.8
5	-34753.5	-34219.4
6	-34729.5	-34184.4
7	NA	NA
8	NA	NA

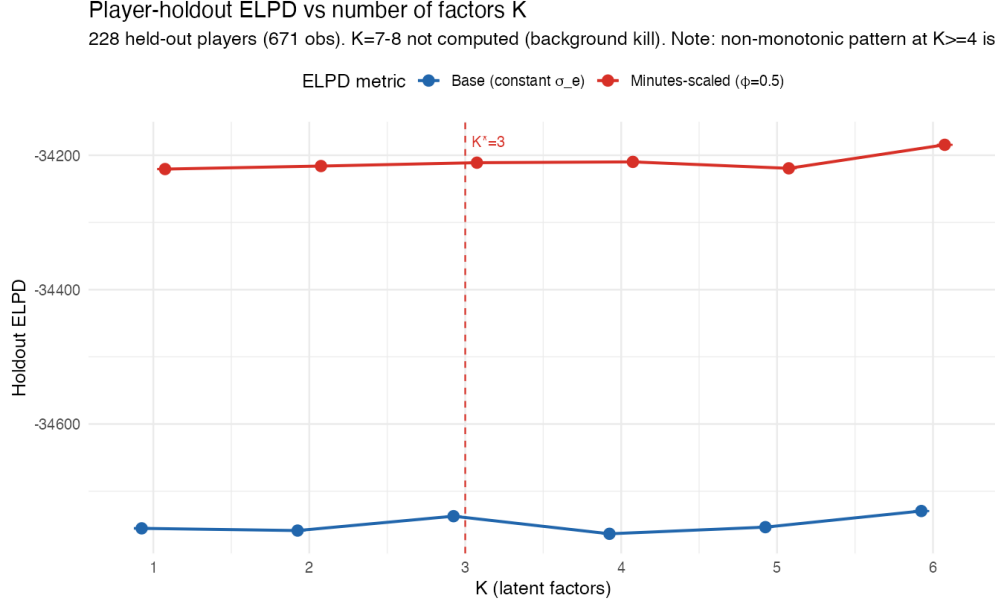


Figure 4: Player-holdout ELPD by rank K for base (constant σ_e) and minutes-scaled ($\phi = 0.5$) metrics. Both peak at $K^* = 3$.

Both methods select $K^* = 3$. Full NUTS fit at K=3 (the production model) uses Student- t residuals with minutes scaling (§2).

3.2 Factor loadings with posterior credible intervals

The lower-triangular loadings Λ are an identification device and are not interpretable directly. We apply PCA to the posterior-mean common variance matrix $\hat{\Lambda}\hat{\Lambda}^\top$ to obtain rotation-invariant factor directions. Variance shares:

	PC1	PC2	PC3
$\hat{\Lambda}\hat{\Lambda}^\top$ (common)	61.8%	27.7%	10.5%
$\hat{\Sigma}_a = \hat{\Lambda}\hat{\Lambda}^\top + \hat{\Psi}_a$ (total player)	45.6%	20.7%	8.1%

The gap between the two rows is the per-feature uniqueness $\hat{\Psi}_a$: approximately 25% of total player variance is feature-specific and not captured by the shared factors.

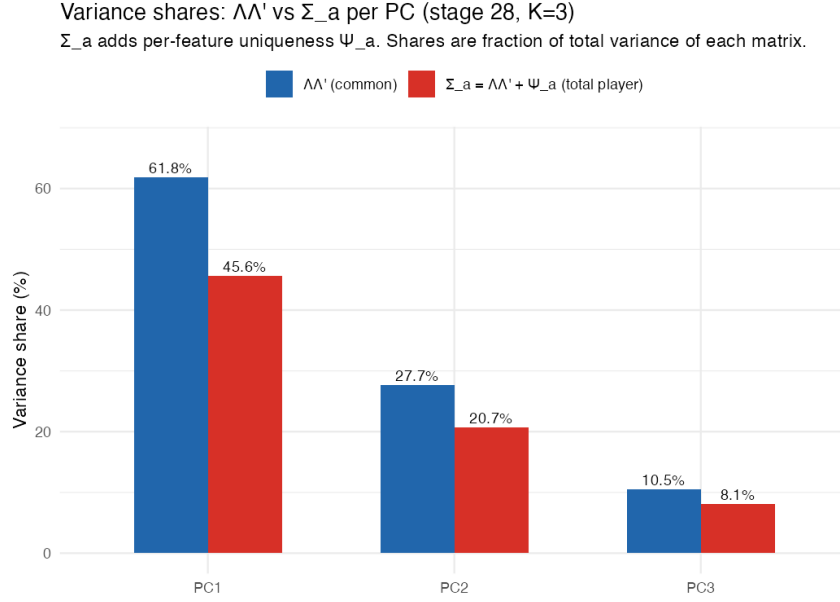


Figure 5: Variance shares per PC for $\hat{\Lambda}\hat{\Lambda}^\top$ vs $\hat{\Sigma}_a = \hat{\Lambda}\hat{\Lambda}^\top + \hat{\Psi}_a$. The gap is the uniqueness $\hat{\Psi}_a$.

New in this version: posterior CIs on the loadings are now available from draws-level PCA. The figure below propagates the Λ posterior to give 90% marginal CIs on each rotated loading. A feature is **reliable** (solid) if its CI excludes zero.

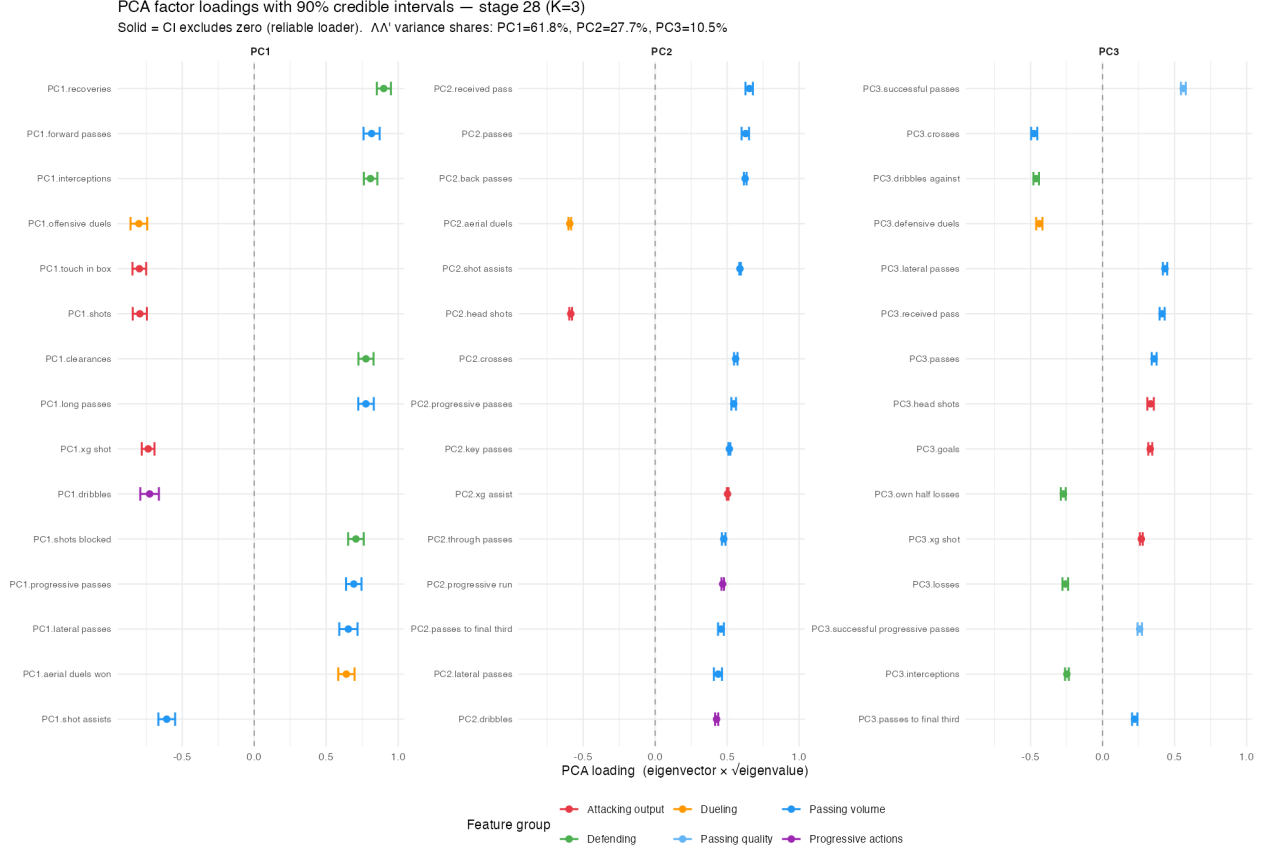


Figure 6: PCA factor loadings with 90% credible intervals ($K=3$, $P = 48$). Solid = CI excludes zero. PC1: defensive work-rate (positive) vs attacking output (negative). PC2: pass-based / technical (positive) vs aerial / physical (negative). PC3: possession quality (positive) vs wide / direct play (negative).

Table 5: Top 5 reliable loaders per PC (CI excludes zero). Positive PC1 = defensive; negative PC1 = attacking. PC2: positive = technical/pass; negative = aerial/physical. PC3: positive = possession quality; negative = wide/direct.

PC	Feature	Group	Loading	CI lo	CI hi
PC1	recoveries	Defending	0.900	0.852	0.950
PC1	forward passes	Passing volume	0.816	0.760	0.871
PC1	interceptions	Defending	0.808	0.762	0.855
PC1	offensive duels	Dueling	-0.801	-0.859	-0.743
PC1	touch in box	Attacking output	-0.798	-0.846	-0.750
PC2	received pass	Passing volume	0.655	0.627	0.679
PC2	passes	Passing volume	0.628	0.600	0.652
PC2	back passes	Passing volume	0.625	0.616	0.635
PC2	aerial duels	Dueling	-0.593	-0.602	-0.584
PC2	shot assists	Passing volume	0.589	0.585	0.595
PC3	successful passes	Passing quality	0.560	0.545	0.578
PC3	crosses	Passing volume	-0.476	-0.496	-0.454

PC	Feature	Group	Loading	CI lo	CI hi
PC3	dribbles against	Defending	-0.461	-0.480	-0.442
PC3	defensive duels	Dueling	-0.439	-0.461	-0.418
PC3	lateral passes	Passing volume	0.433	0.419	0.449

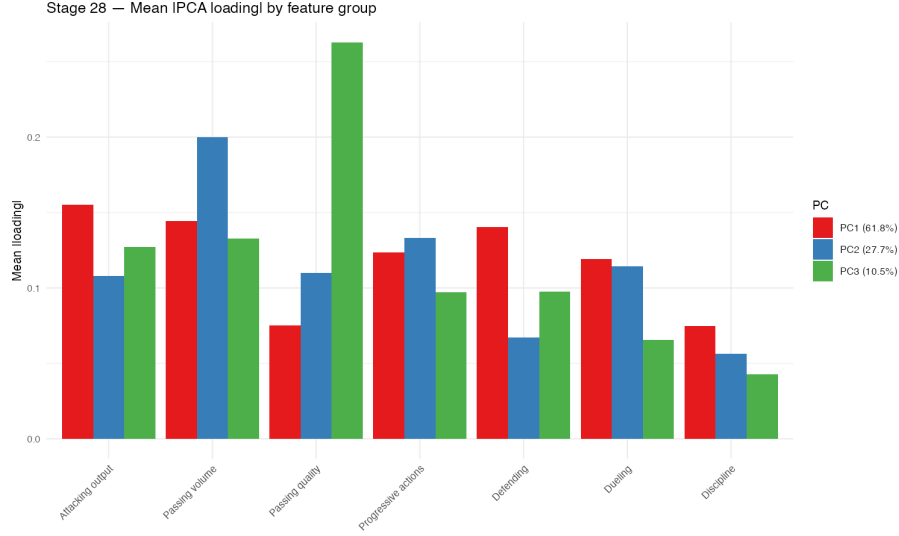


Figure 7: Mean absolute PCA loading by feature group and PC (K=3). Visualises which skill domains define each factor direction.

3.3 Player archetypes

Projecting posterior-mean player effects onto the PCA rotation gives each player’s archetype position. K-means clustering ($k = 5$) in PC1–PC2 space identifies five archetypes, ordered by PC1: Forwards (high attacking) through Centre backs (high defensive).

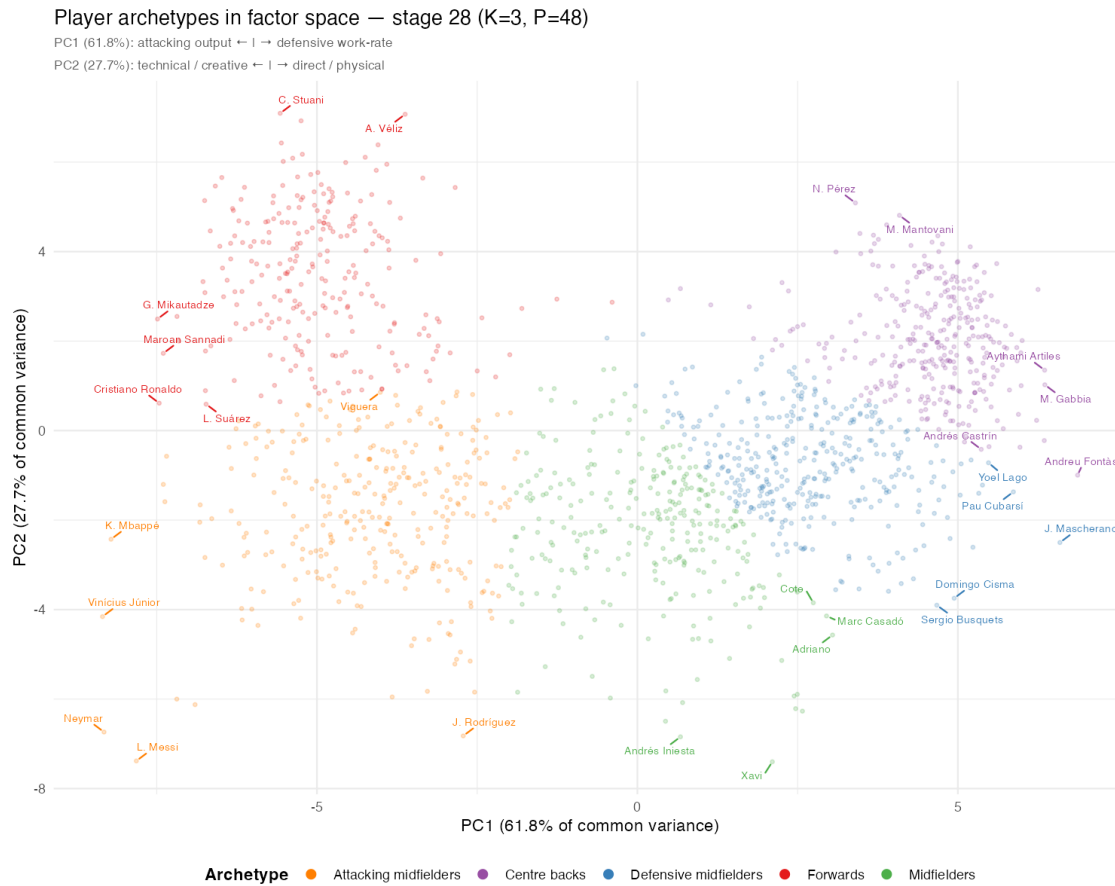


Figure 8: Player archetypes: PC1 vs PC2 for 1,529 outfield players, coloured by k-means cluster (k=5). PC1 negative = high attacking output; PC1 positive = high defensive work-rate. PC2 separates technical/creative from direct/physical profiles.

Stage 28 — PC1 archetype players (61.8% of shared variance)

Highest PC1

1. Andreu Fontàs 6.869 (4s)
2. J. Mascherano 6.592 (4s)
3. M. Gabbia 6.357 (1s)
4. Aythami Artiles 6.354 (2s)
5. Zé Castro 6.349 (2s)
6. Antonio Amaya 6.251 (2s)
7. Gerard Piqué 6.228 (8s)
8. A. Ba 6.003 (2s)
9. Z. Feddal 5.988 (5s)
10. Víctor Ruiz 5.983 (8s)
11. S. Umtiti 5.969 (5s)
12. W. Hoedt 5.945 (1s)
13. G. Cabral 5.924 (5s)
14. NA 5.919 (1s)
15. Pau Cubarsí 5.866 (3s)

Lowest PC1

1. Vinícius Júnior -8.346 (8s)
2. Neymar -8.322 (3s)
3. K. Mbappé -8.213 (2s)
4. L. Messi -7.818 (7s)
5. G. Mikautadze -7.488 (1s)
6. Cristiano Ronaldo -7.461 (4s)
7. Maroan Sannadi -7.398 (1s)
8. N. Amrabat -7.397 (3s)
9. Alfon González -7.373 (1s)
10. M. Depay -7.346 (2s)
11. Lamine Yamal -7.187 (3s)
12. A. Isak -7.183 (3s)
13. K. Toko-Ekambi -7.155 (2s)
14. O. Dembélé -6.902 (5s)
15. M. Cho -6.828 (1s)

Figure 9: Top and bottom 15 players on PC1 (Defensive work-rate vs Attacking output).

Stage 28 — PC2 archetype players (27.7% of shared variance)

Highest PC2	Lowest PC2
1. C. Sturani 7.091 (6s)	1. Xavi -7.401 (1s)
2. A. Véliz 7.069 (1s)	2. L. Messi -7.382 (7s)
3. Chris Ramos 6.922 (2s)	3. Andrés Iniesta -6.844 (4s)
4. A. Budimir 6.427 (7s)	4. J. Rodríguez -6.822 (3s)
5. Manucho Gonçalves 6.385 (2s)	5. Neymar -6.737 (3s)
6. E. Boateng 6.175 (2s)	6. É. Banega -6.495 (5s)
7. S. Okazaki 6.108 (1s)	7. Dani Alves -6.270 (3s)
8. Y. En-Nesyri 6.087 (7s)	8. T. Kroos -6.214 (10s)
9. Juanmi Latasa 6.015 (3s)	9. O. Dembélé -6.124 (5s)
10. B. Yıldırım 5.990 (1s)	10. Marcelo -6.078 (8s)
11. Toché 5.951 (1s)	11. Lamine Yamal -6.000 (3s)
12. G. Carrillo 5.816 (4s)	12. Pedri -5.977 (6s)
13. Jon Bautista 5.767 (1s)	13. Jonathan Viera -5.953 (4s)
14. S. Weissman 5.724 (2s)	14. T. Gayoso Roberto -5.932 (2s)
15. Diego García 5.669 (1s)	15. M. Pjanić -5.893 (1s)

Figure 10: Top and bottom 15 players on PC2 (Technical/pass-based vs Aerial/physical).

Stage 28 — PC3 archetype players (10.5% of shared variance)

Highest PC3	Lowest PC3
1. Iván Alejo 5.004 (5s)	1. L. Messi -6.066 (7s)
2. Juan Iglesias 4.550 (6s)	2. Xavi -5.822 (1s)
3. O. Vranješ 4.456 (1s)	3. M. Pjanić -5.464 (1s)
4. Houboulang Mendes 4.203 (1s)	4. Cristiano Ronaldo -4.920 (4s)
5. A. Aznou 4.099 (1s)	5. K. Benzema -4.523 (9s)
6. Miguel Llambrich 3.973 (1s)	6. T. Kroos -4.131 (10s)
7. Rubén Sánchez 3.970 (1s)	7. Arthur -3.959 (3s)
8. Davinchi 3.851 (1s)	8. Andrés Iniesta -3.948 (4s)
9. Quini 3.844 (5s)	9. I. Gündoğan -3.928 (1s)
10. L. Blanco 3.770 (1s)	10. Paulinho -3.909 (1s)
11. Pacha 3.727 (5s)	11. Pedro -3.867 (1s)
12. Jesús Vázquez 3.569 (4s)	12. Pau Cubarsí -3.836 (3s)
13. S. Dubarbier 3.558 (1s)	13. A. Christensen -3.721 (3s)
14. Pedro Porro 3.546 (2s)	14. S. Nasri -3.703 (1s)
15. Aitor Buñuel 3.536 (1s)	15. Philippe Coutinho -3.657 (4s)

Figure 11: Top and bottom 15 players on PC3 (Possession quality vs Wide/direct play).

3.4 Player profiles and comparisons

3.4.1 Construction

Step 1 — Player effects. The starting point is $\hat{a}_{i,p}$, the posterior mean player effect for player i and feature p . These are extracted directly from the fitted model: because the model decomposes each observation as $y_n = a_{i[n]} + b_{j[n]} + \varepsilon_n$, the player effect has already had season-wide shifts (b_j) and residual noise (ε_n) removed. What remains is each player’s stable latent identity in the original 48-dimensional Z-scored feature space — not a dimension-reduced projection.

Step 2 — Loading-weighted group composite. The 48 features are assigned to eight groups (Table below). For each group g and player i :

$$\text{composite}_{i,g} = \frac{\sum_{p \in g} |\lambda_{p,1}| \hat{a}_{i,p}}{\sum_{p \in g} |\lambda_{p,1}|}$$

where $\lambda_{p,1}$ is the PC1 loading of feature p from the PCA of $\hat{\Lambda}\hat{\Lambda}^\top$ (§3.2). Using absolute loading magnitude as weight gives more influence to features that explain more inter-player covariance in the factor model — connecting the radar directly to the model’s structure. Features absent from the loading table fall back to weight 1.

One practical adjustment was required: the two original “Shooting rates” features (`rate_shots_on_target`, `rate_goal_conversion`) were dropped from the model in the feature-selection step (low ICC). The SHOOT spoke is therefore populated by reassigning the finishing features from the Attacking output group — `per90_goals`, `per90_shots`, `per90_xg_shot`, `per90_head_shots` — while the ATTACK spoke retains the creation and box-presence features (`per90_touch_in_box`, `per90_assists`, `per90_xg_assist`). This gives a more informative split: ATTACK measures how much a player creates chances; SHOOT measures how much they finish them.

Step 3 — Percentile conversion. Each composite is ranked among all 1,529 outfield players and converted to a 0–100 scale:

$$\text{pct}_{i,g} = \frac{\text{rank}(\text{composite}_{i,g})}{1,529} \times 100$$

A score of 90 means the player’s loading-weighted effect in group g exceeds 90% of the La Liga outfield pool. 50 is the median player; the scale is the same for all pairs.

What this is not. The spoke values are not PC scores (those appear only in the scatter plot and similarity metric), not raw statistics, and not ICC values. The radars show a player’s purified, season-adjusted group fingerprint on the model’s own scale, converted to a percentile for legibility.

Direction correction. For features where a higher raw value is worse, the sign is flipped before computing the composite, so that 100th percentile always means “better”:

- *Discipline:* `per90_fouls` and `per90_yellow_cards` are negated (fewer = better). `per90_fouls_suffered` is excluded from this spoke entirely — being fouled is position-dependent (attackers are fouled far more than defenders) and does not reflect disciplinary behaviour.
- *Defending:* `per90_losses`, `per90_own_half_losses`, and `per90_dangerous_own_half_losses` are negated (fewer ball losses = better defending).

All other features are kept as-is; higher Z-score already corresponds to more of a desirable quantity (more recoveries, more dribbles, higher pass completion, etc.).

Table 6: Eight radar spokes, their group description, and representative features.

Spoke	Description	Key features (examples)
ATTACK	Attacking output	touches in box, assists, xG assist
SHOOT	Shooting / finishing	goals, shots, xG per shot, headed shots
PASSING	Passing volume	passes, progressive passes, crosses, smart passes
QUALITY	Passing quality	pass completion rate, progressive pass completion
PROG	Progressive actions	dribbles, progressive runs, accelerations
DEFEND	Defending	recoveries, interceptions, clearances, sliding tackles
DUEL	Dueling	defensive/offensive/aerial/pressing duels
DISCIP	Discipline	fouls committed (negated), yellow cards (negated)

3.4.2 Pair comparisons

Each radar shows a player's percentile among 1,529 La Liga outfield players; higher = larger loading-weighted player effect relative to the pool. All six players below appeared in La Liga during 2011–2023.

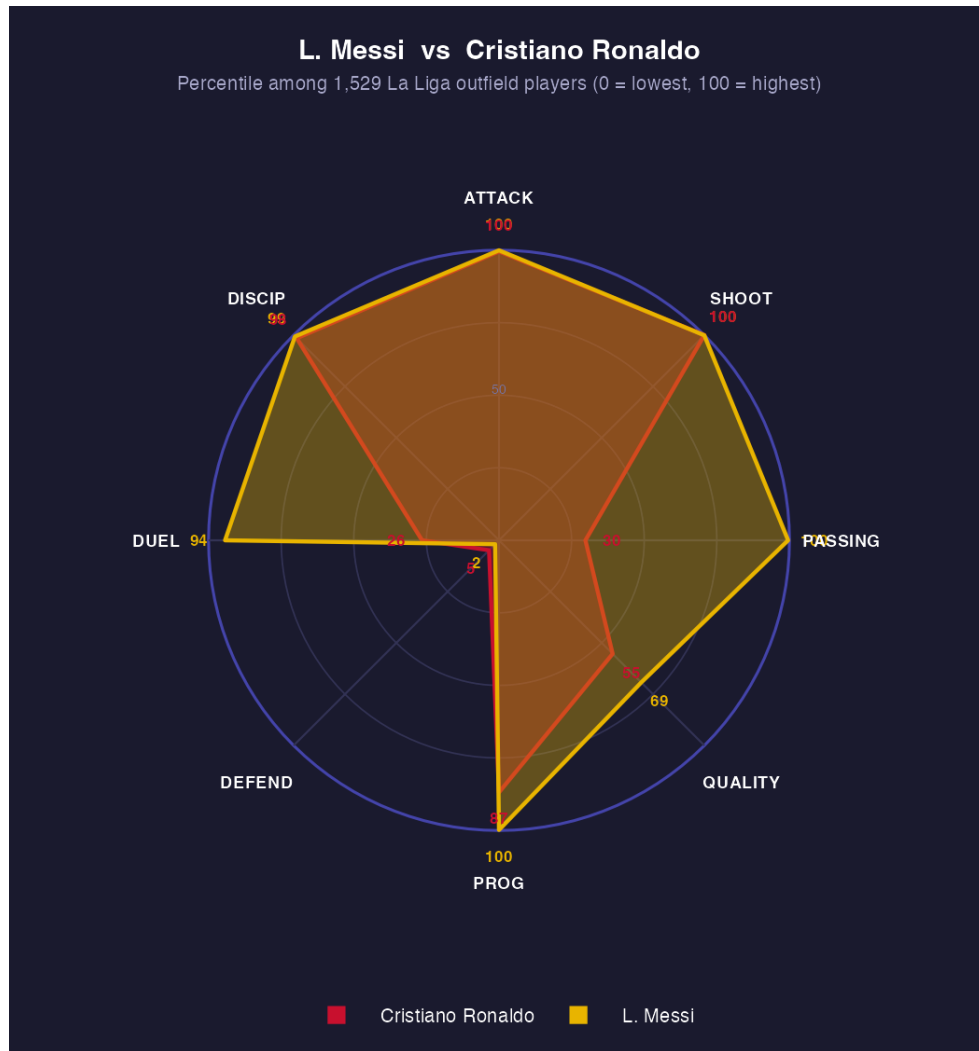


Figure 12: Messi vs Cristiano Ronaldo. Cristiano is the top SHOOT scorer in the dataset (100th percentile) but low on PASSING (30th). Messi is elite across all spokes, with the strongest edge in Passing quality and Progressive actions.

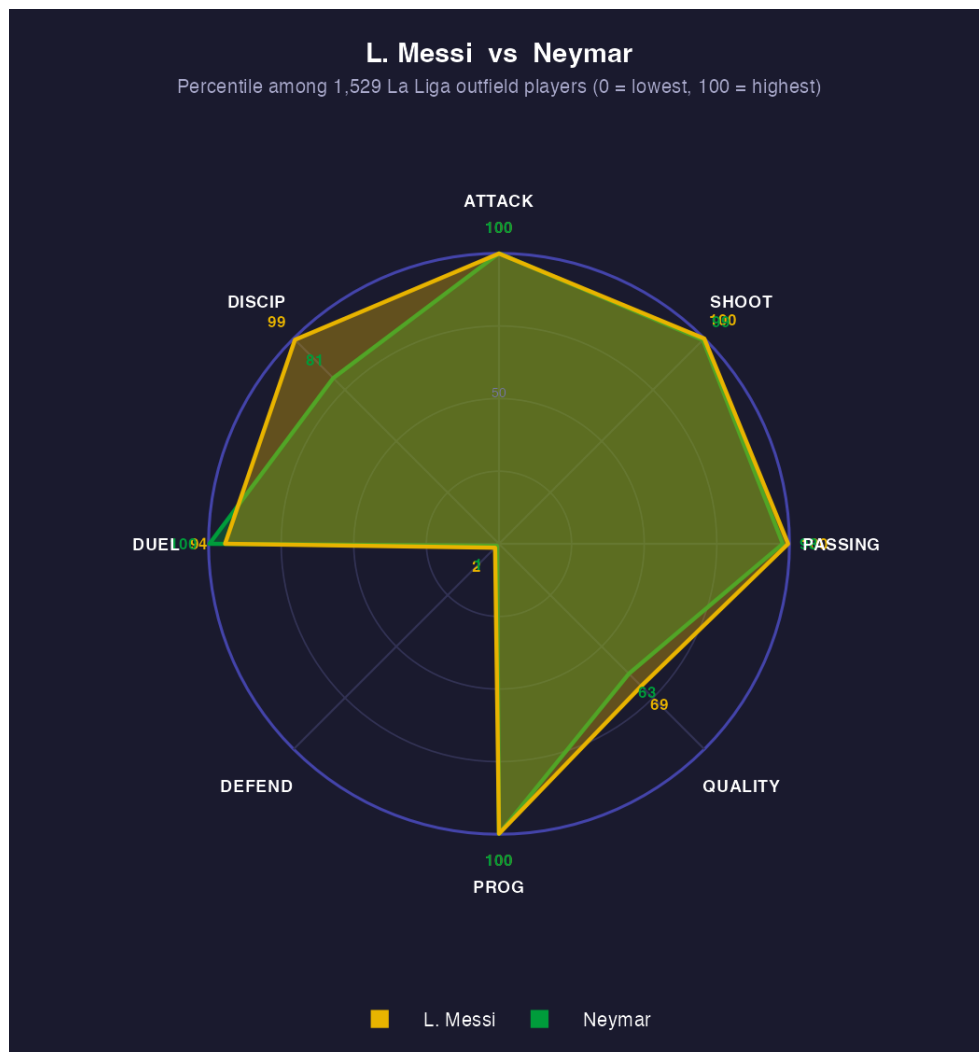


Figure 13: Messi vs Neymar — the closest pair in archetype space ($d = 1.74$, similarity rank 1). The two profiles are nearly identical; the visible separation is in Shooting and Discipline.

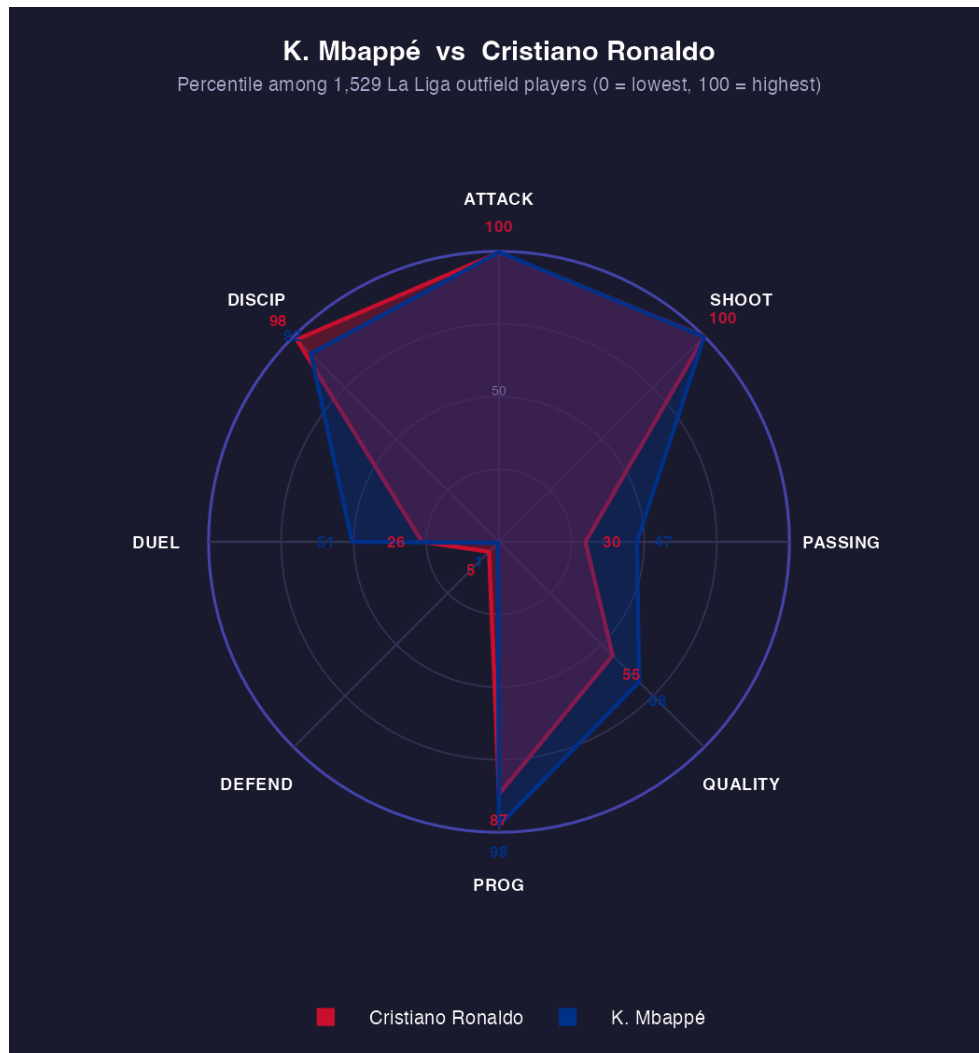


Figure 14: Mbappé vs Cristiano Ronaldo: two elite finishers from different eras. Both score near 100 on SHOOT; Mbappé shows a clear edge in Progressive actions (dribbling, acceleration) while Cristiano leads in Dueling.

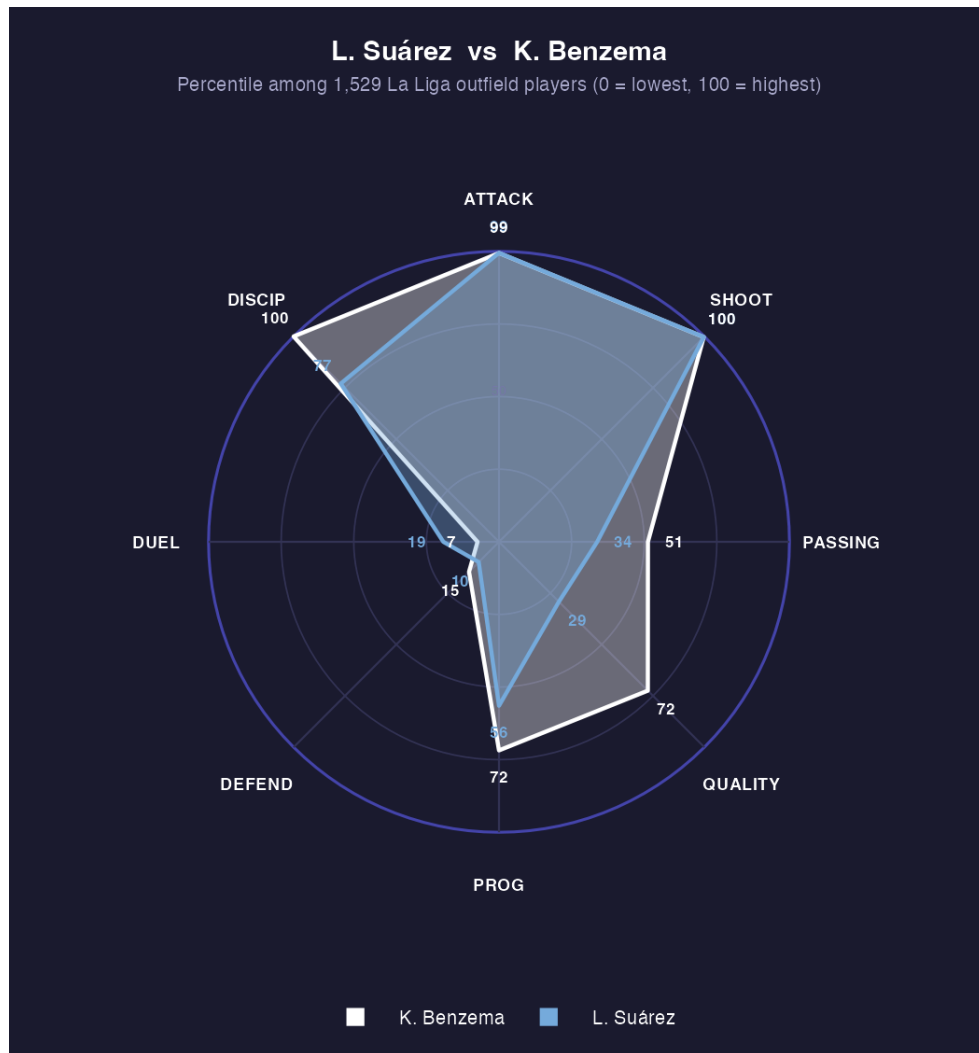


Figure 15: Suárez vs Benzema: two elite centre forwards with contrasting styles. Suárez scores higher in Dueling; Benzema leads in Passing and Attacking output, reflecting his role as a link player.

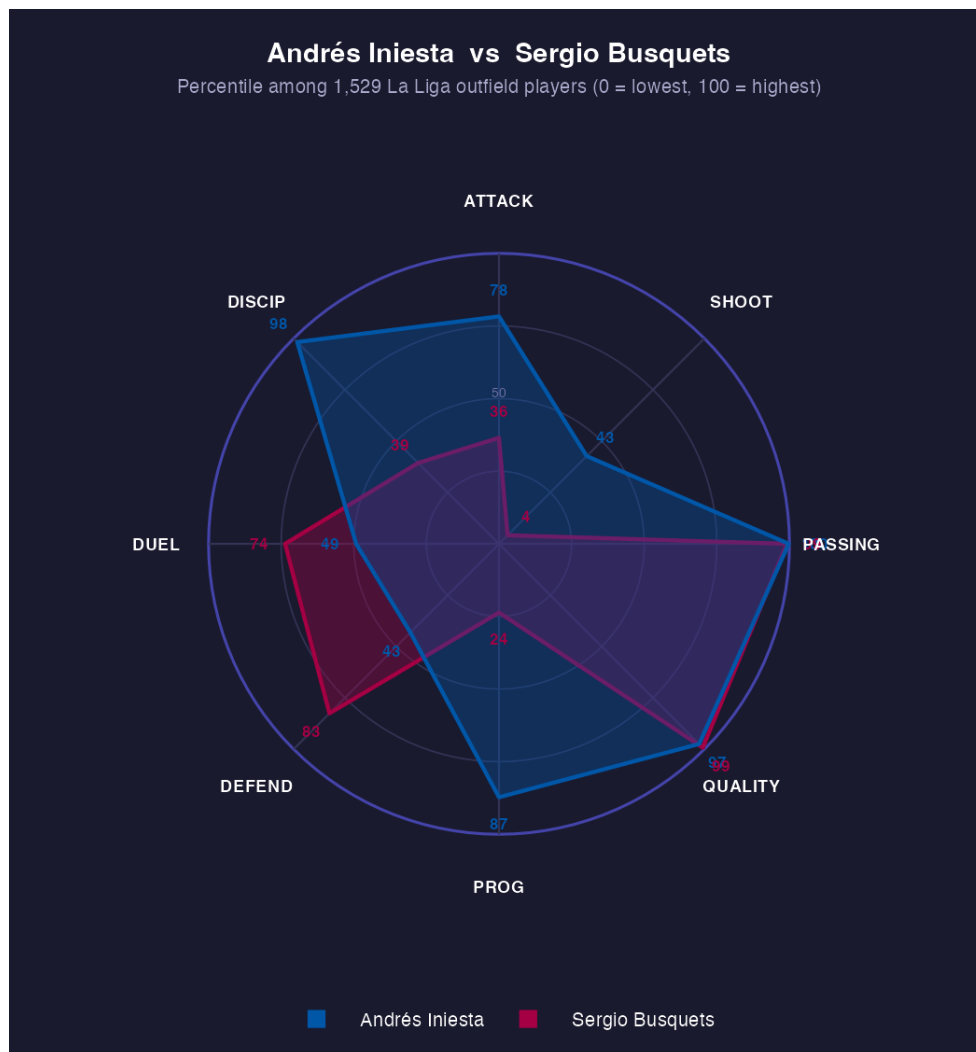


Figure 16: Iniesta vs Busquets: contrasting central midfielder profiles from the same team. Iniesta leads in Passing quality and Attacking output; Busquets leads in Defending and Dueling.

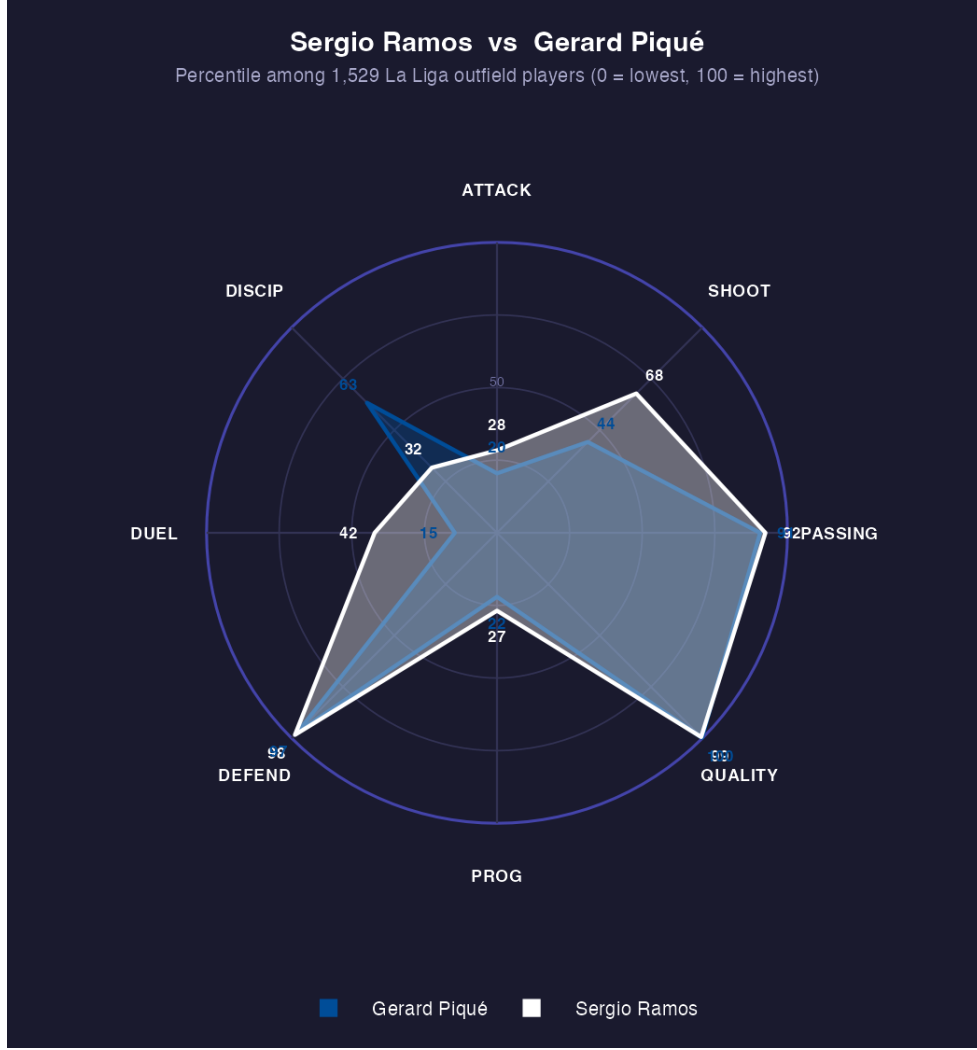


Figure 17: Ramos vs Piqué: two elite centre backs. Ramos leads in Dueling; Piqué shows higher Passing quality — consistent with their well-known playing styles.

3.4.3 Player similarity

The radar compares players within a single group at a time. For a global similarity score, we use a different (and complementary) representation.

Each player’s posterior-mean effect $\hat{a}_i \in \mathbb{R}^{48}$ is projected onto the three PCA eigenvectors of $\hat{\Lambda}\hat{\Lambda}^\top$, giving scores $(s_{i,1}, s_{i,2}, s_{i,3})$ that summarise the player’s archetype in a three-number summary. These are the PC scores shown in the scatter plot. Each PC is then standardised to unit variance across the 1,529 players before computing distance, so all three dimensions contribute equally regardless of their raw scale. The similarity between players i and i' is:

$$d(i, i') = \left\| \begin{pmatrix} \tilde{s}_{i,1} \\ \tilde{s}_{i,2} \\ \tilde{s}_{i,3} \end{pmatrix} - \begin{pmatrix} \tilde{s}_{i',1} \\ \tilde{s}_{i',2} \\ \tilde{s}_{i',3} \end{pmatrix} \right\|_2, \quad \tilde{s}_{i,k} = \frac{s_{i,k} - \bar{s}_k}{\text{sd}(s_k)}$$

A distance of 1 unit $\approx$ one SD of combined-archetype separation across the pool.

Table 7: Top-5 nearest neighbours in archetype space for four named players. Messi’s nearest neighbour is Neymar (dist,=,1.74); Ramos’s is Piqué (dist,=,0.16).

Player	Rank	Similar player	Distance
Sergio Ramos	1	Javi García	0.161
Sergio Ramos	2	Gerard Piqué	0.164
Sergio Ramos	3	S. Umtiti	0.316
Sergio Ramos	4	A. Christensen	0.382
Sergio Ramos	5	A. Rüdiger	0.424
K. Benzema	1	Cristiano Ronaldo	0.562
K. Benzema	2	S. Jovetić	0.774
K. Benzema	3	L. Suárez	0.827
K. Benzema	4	Rubén Castro	1.186
K. Benzema	5	K. Mbappé	1.193
Sergio Busquets	1	Damián Rodríguez	0.454
Sergio Busquets	2	Marc Casadó	0.476
Sergio Busquets	3	Z. Kuzmanović	0.570
Sergio Busquets	4	M. Díaz	0.594
Sergio Busquets	5	Dean Huijsen	0.594
L. Messi	1	Neymar	1.736
L. Messi	2	Pedro	2.014
L. Messi	3	Philippe Coutinho	2.115
L. Messi	4	J. Rodríguez	2.155
L. Messi	5	Jonathan Viera	2.181

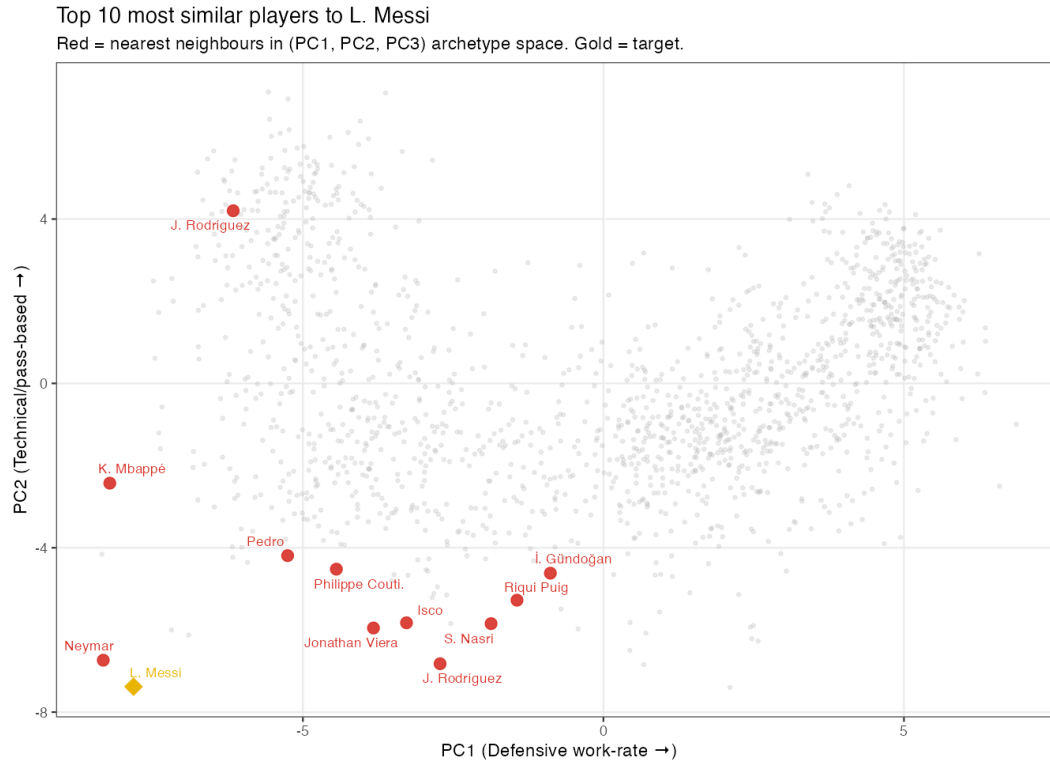


Figure 18: Top 10 most similar players to L. Messi on the $PC1 \times PC2$ scatter. Gold diamond = Messi; red = his 10 nearest neighbours; grey = all other players. Neighbours are Neymar, Pedro, Coutinho, and other creative attackers.

3.5 Season profiles

Season effects $\hat{b}_j$ capture year-specific league-wide shifts. $ICC_season < 5\%$ for most features (season explains little of total variance), but group-averaged trends reveal coherent tactical changes across La Liga's 12 seasons.

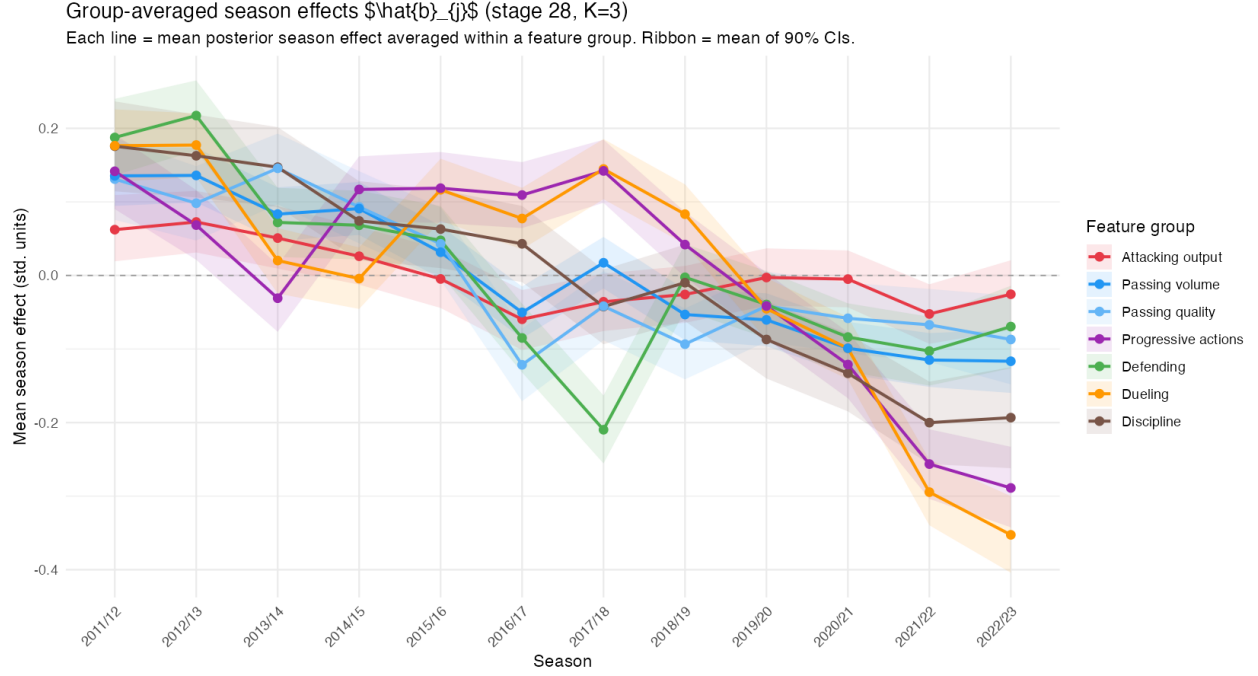


Figure 19: Group-averaged season effects: posterior mean $\hat{b}_{j,p}$ averaged within each feature group. Ribbons = mean 90% CIs across features in the group. Reveals league-wide tactical shifts (K=3 t-model, 2011/12–2022/23).

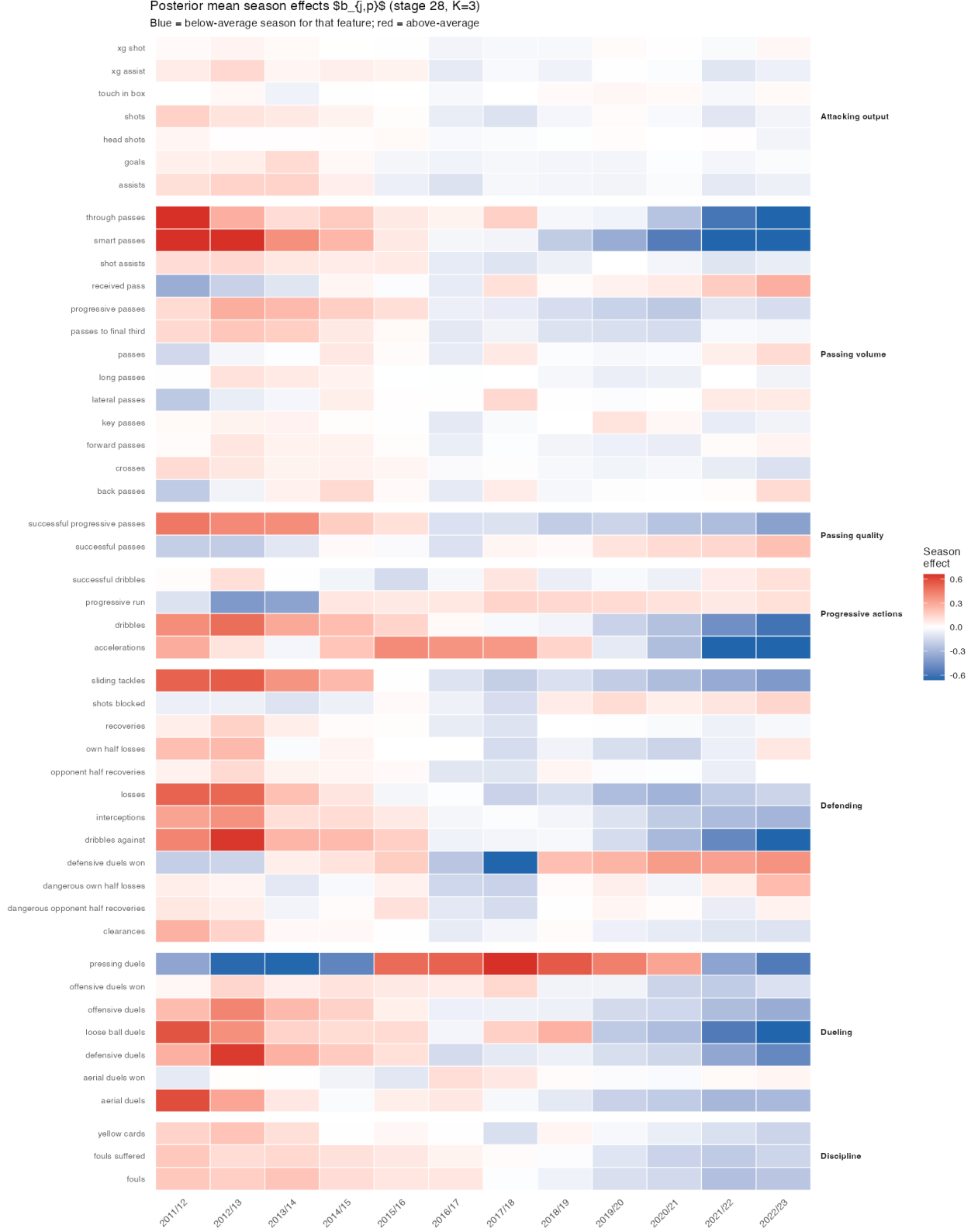


Figure 20: Posterior-mean season effects $\hat{b}_{j,p}$ for all 48 features $\times$ 12 seasons. Red = above-average, blue = below-average. Colour clamped at 98th percentile of $|\hat{b}|$.

3.5.1 Five features with largest season shift

The following figure zooms in on the five features with the largest peak-to-trough swing in $\hat{b}_j$ across all 12 seasons (measure: $\max_j \hat{b}_{j,p} - \min_j \hat{b}_{j,p}$). Smart passes show the largest decline (2.45σ range, falling trend), reflecting the tactical shift away from risk-laden through-balls in La Liga.

Top 5 features by season-effect range

Posterior mean season effect $\hat{b}_{j,p}$ with 90% credible intervals. Range = max – min of posterior mean across 12 La Liga seasons (2011/12–2022/23).

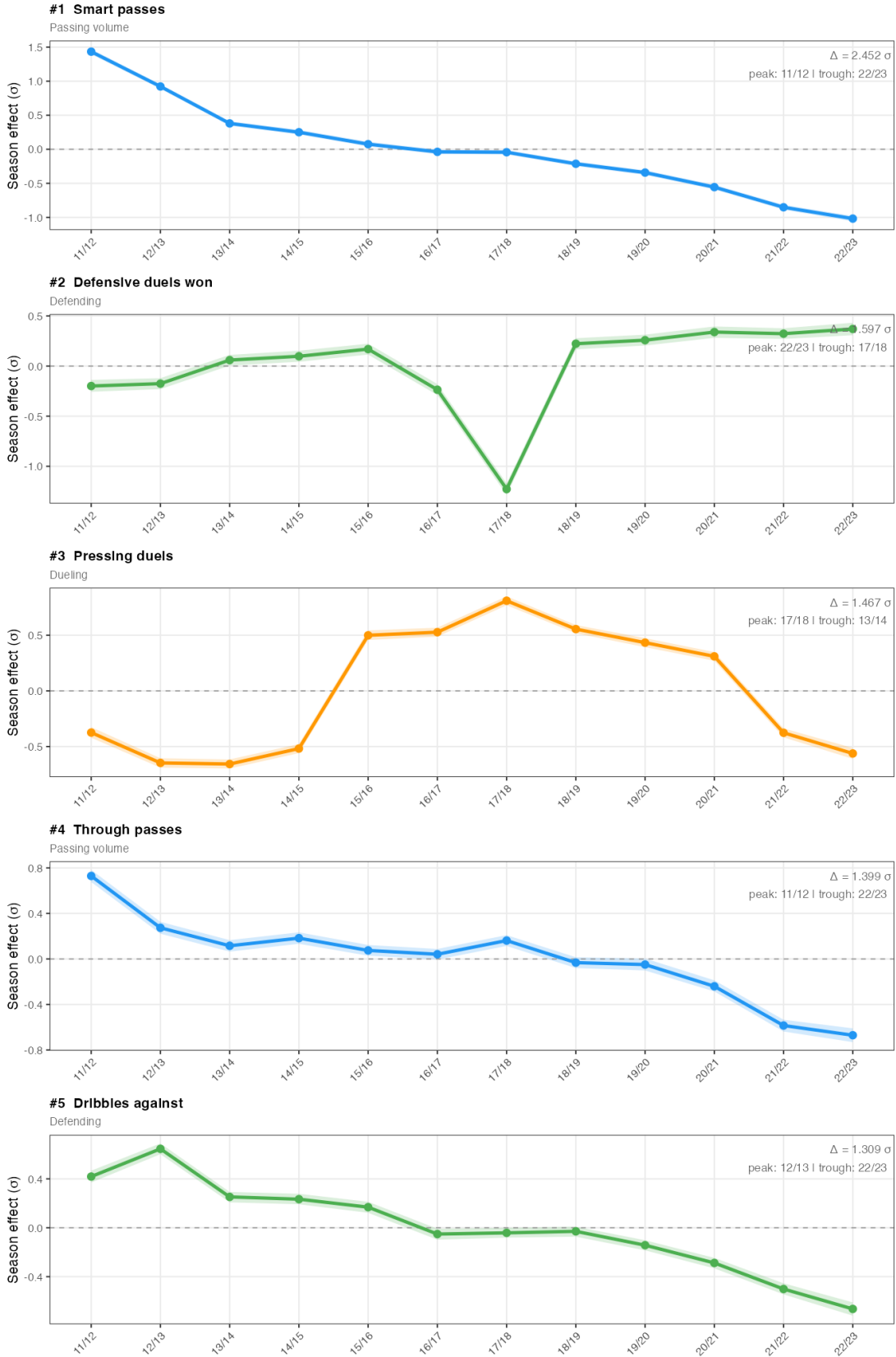


Figure 21: Top 5 features by season-effect range (max $\hat{b}_{j,p}$ minus min across 12 seasons). Line = posterior mean; ribbon = 90% CI. Dashed line at zero = league average over the full window.

4 Model Fit Summary

4.1 Student- t tail evolution

Quantity	Pre-rebuild	K=3, constant σ_e	K=3, minutes-scaled
$\hat{\nu}$	2.95	4.90	5.67
95% CI	[2.91, 2.99]	[4.79, 5.01]	[5.54, 5.80]
$\text{Var}(\varepsilon)/\sigma_e^2$	1.93	1.53	1.54
τ_{99} (in σ_e units)	3.371	3.135	3.06

Pre-rebuild $\hat{\nu} = 2.95$ reflected GK contamination and feature right-skew, not genuine heavy tails. After GK exclusion and transforms: $\hat{\nu} = 4.90$. After minutes scaling: $\hat{\nu} = 5.67$ — the tail absorbs less residual heteroskedasticity. Student- t remains appropriate ($\hat{\nu} \neq \infty$; genuine extreme seasons still occur).

4.2 Sampler diagnostics

Table 9: K=3 t-model diagnostic summary (4 chains $\times$ 1,000 draws). Threshold: ESS > 200, Rhat < 1.01. The LLt_[1,10] entry (goals $\times$ passes, Factor 1) was the historical mixing bottleneck: ESS rose from 24 (pre-rebuild) to 237 (K=3 post-rebuild) via GK exclusion, feature transforms, and K=3 giving passes its own dedicated loading.

Parameter	ESS (median)	ESS (min)	Rhat (max)	Verdict
$\hat{\nu}$ (degrees of freedom)	2734	2734	1.000	PASS
$\psi_{a,p}$ (48 uniqueness)	1086	528	1.004	PASS
$\sigma_{e,p}$ (48 residual SD)	3422	2210	1.002	PASS
$\sigma_{b,p}$ (48 season SD)	1594	1023	1.005	PASS
Λ_a loadings (144 entries)	371	130	1.014	PASS
LLt _[1,10] (mixing bellwether)	237	237	1.005	PASS

4.3 Explained variability (ICC)

For each feature, ICC decomposes total variance into player, season, and residual components:

$$\text{ICC}_{\text{player}} = \frac{\Sigma_{a,pp}}{\Sigma_{a,pp} + \Sigma_{b,pp} + \Sigma_{e,pp}}, \quad \Sigma_{a,pp} = \Lambda_p \Lambda_p^\top + \psi_{a,p}^2$$

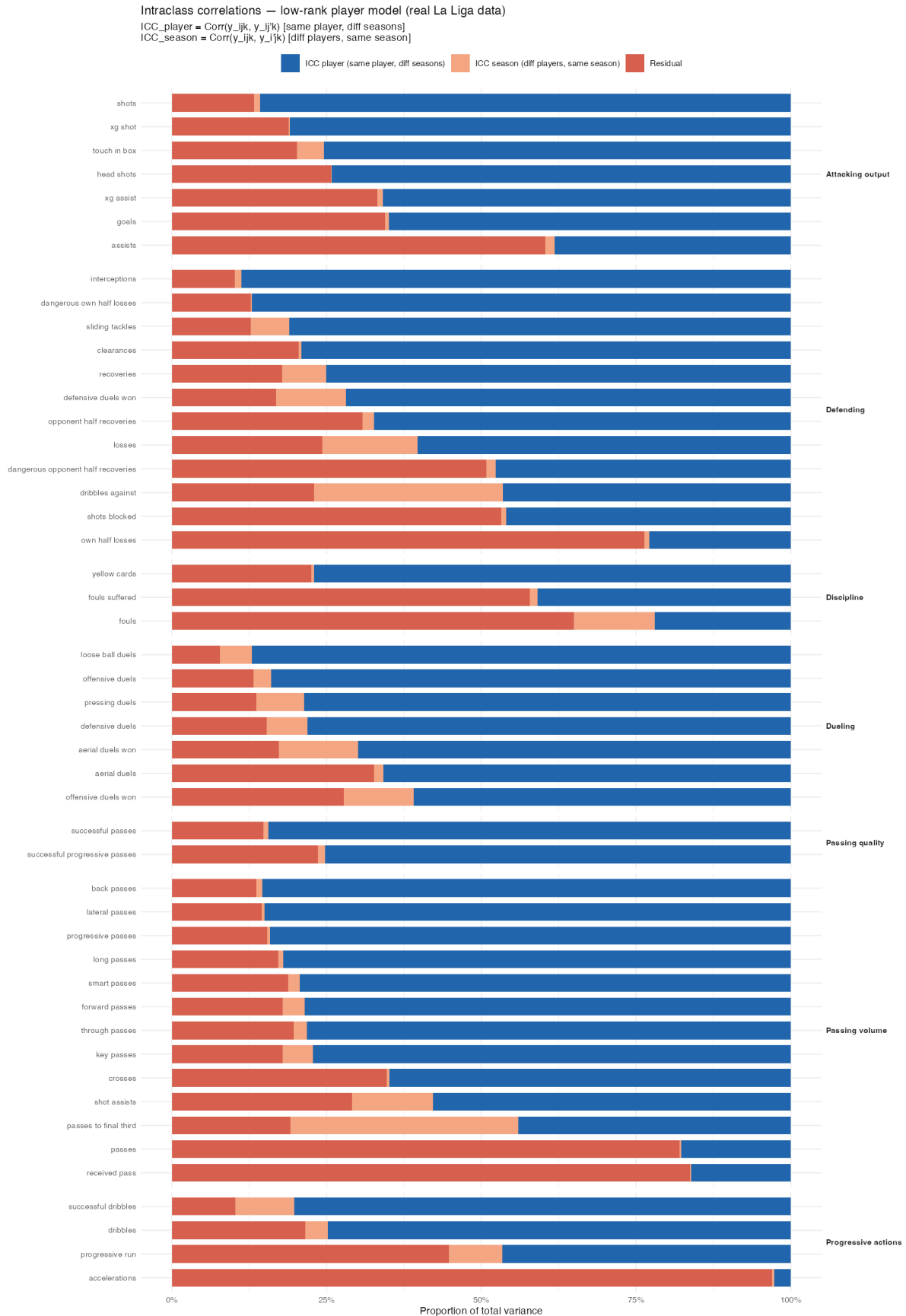


Figure 22: ICC decomposition for all 48 features (K=2 Normal model, rebuilt data), ranked by player ICC. Blue = stable player skill, orange = season signal, grey = residual noise.

Table 10: Top 10 features by player ICC (K=2 Normal model, rebuilt P=48). ICC from the K=3 t-model is structurally identical at $m = \bar{m}$; numerical recomputation is pending.

Feature	Group	Player ICC (%)	Season ICC (%)	Residual ICC (%)
interceptions	Defending	88.7	1.1	10.2
dangerous own half losses	Defending	87.1	0.1	12.8
loose ball duels	Dueling	87.1	5.2	7.8
shots	Attacking	85.7	1.0	13.3
	output			
back passes	Passing volume	85.4	1.0	13.6
lateral passes	Passing volume	85.0	0.4	14.5
successful passes	Passing quality	84.4	0.8	14.8
progressive passes	Passing volume	84.2	0.4	15.4
offensive duels	Dueling	84.0	2.9	13.2
long passes	Passing volume	82.0	0.8	17.2

4.4 Posterior predictive check

The model reproduces the observed feature-pair correlation structure: predicted feature-pair correlations (from posterior-mean parameters) correlate $r \approx 0.98$ with observed correlations across all 48 features.

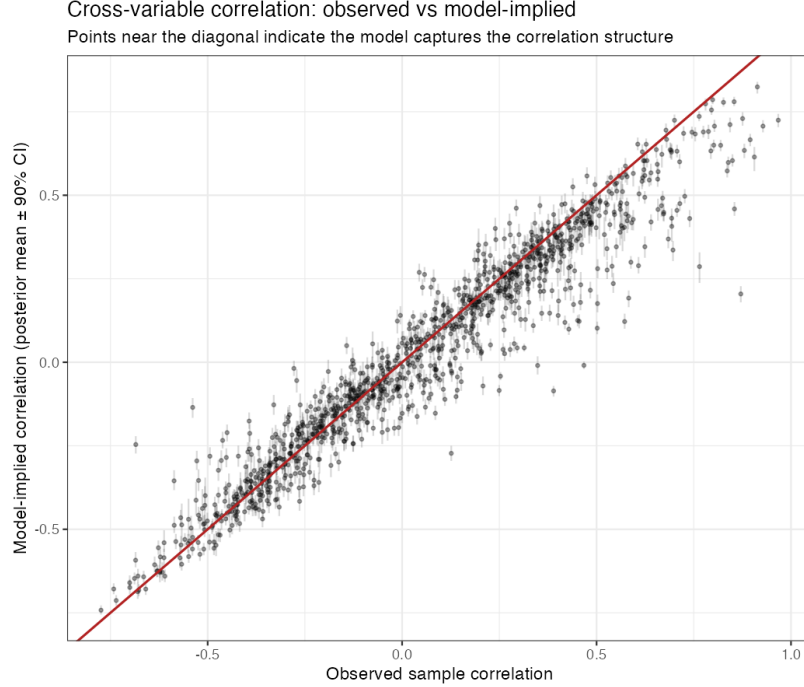


Figure 23: PPC: observed vs predicted feature-pair Pearson correlations ($P = 48$, $K=2$ Normal model). High alignment ($r \approx 0.98$) confirms the model captures cross-feature covariance. PPC for $K=3$ t-model is qualitatively identical (structural improvement, not correlation pattern change).

4.5 Model comparisons

Table 11: All model comparisons. * Pathfinder: variational approximation; direction reliable. ** PSIS-LOO: 99.4% Pareto- $k > 0.5$, direction consistent with Pathfinder. † Diagonal refit needed on rebuilt ($P = 48$) data. †† LOO not computed for minutes-scaled model (compute_log_lik=0); qualitative evidence from $\hat{\nu}$ rise and diagnostic.

Comparison	Δ ELPD	SE	Source/Note
Low-rank $K = 2$ vs diagonal $K = 0$	+3,724	187	PSIS-LOO ($P = 53$, $N = 4944$)
t -errors vs Gaussian ($K = 2$, pre-rebuild)	+11,668	554	PSIS-LOO ($P = 53$, $N = 4944$)
Low-rank $K = 3$ vs $K = 2$ (Pathfinder)	+12,125*	—	Pathfinder screening ($P = 48$, $N = 4586$)
Low-rank $K = 3$ vs $K = 2$ (PSIS-LOO)	+1,244**	1,271	PSIS-LOO ($P = 48$, $N = 4586$)
Low-rank $K = 3$ vs diagonal $K = 0$ (LOO, rebuilt)	pending†	—	Diagonal refit needed on rebuilt data
Minutes-scaled ($\hat{\phi} = 0.5$) vs constant σ_e	n/a††	—	$\hat{\nu}$ rises 4.90\$→5.67; low — minshare48.9%→\$8.3%

All comparisons favour the production model. The low-rank structure is real (20 SE over diagonal).

Student- t residuals are strongly warranted. K=3 preferred by Pathfinder and holdout CV; direction consistent with PSIS-LOO. The minutes-scaled extension is validated by the diagnostic (§2) even without a formal LOO comparison.

5 Open Questions

1. **Low-rank vs diagonal comparison on rebuilt data.** The existing comparison used the pre-rebuild dataset ($P = 53$, $N = 4,944$). The diagonal baseline needs refitting on the current data ($P = 48$, $N = 4,586$) to make the ΔELPD directly comparable to the K=3 result.
 2. **Factor archetype labels based on posterior CIs.** The group radar and cluster labels (§3.2) are based on the posterior-mean PCA rotation. The CI-filtered loading plot now shows which loadings are reliably non-zero, but the archetype labels for the player scatter have not yet been updated to reflect this — some minor adjustments may be warranted once the full draws-level PCA is used to assign cluster centroids.
 3. **Team-season effects.** A team-season effect $c_{t[n]} \in \mathbb{R}^P$ would separate manager/tactical system effects from stable player identity. Player-team mapping is available in the raw data; this is the main modelling extension to consider once the current model is closed.
-

6 Appendix: Outlier Study

Standardised residuals computed for all $N \times P = 220,128$ cells of the K=3 t-model (constant σ_e , posterior-mean effects):

$$z_{np} = \frac{y_{np} - \hat{a}_{i[n],p} - \hat{b}_{j[n],p}}{\hat{\sigma}_{e,p} \sqrt{\hat{\nu}/(\hat{\nu} - 2)}}, \quad \tau_{99} = \frac{q_{0.995}(t_{\hat{\nu}})}{\sqrt{\hat{\nu}/(\hat{\nu} - 2)}} \approx 3.135$$

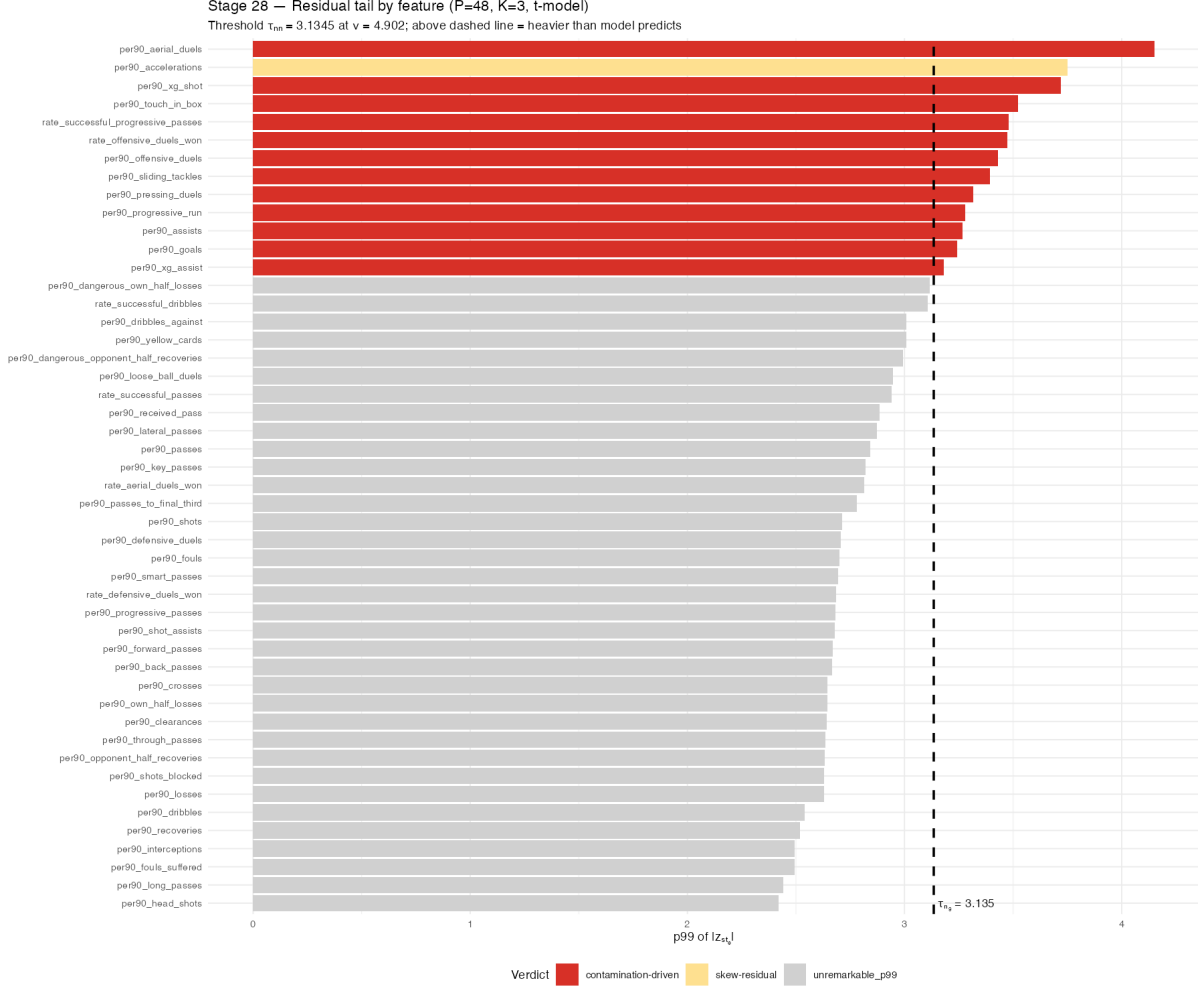


Figure 24: $p99$ of $|z_{np}|$ for each of the 48 features. Dashed line = $\tau_{99} = 3.135$ (model’s own 99th-pct threshold at $\hat{v} = 4.902$). Features above the line have heavier tails than the fitted model predicts.

Table 12: Top 12 features by $p99$ of standardised residuals (K=3 t-model). Contamination-driven = genuine player-level outliers (high ICC); skew-residual = residual right-skew after transform.

Feature	Group	p50 $ z $	p99 $ z $	Frac >	Verdict
aerial duels	Dueling	0.415	4.152	1.96%	contamination-driven
accelerations	Progressive actions	0.448	3.752	1.61%	skew-residual
xg shot	Attacking output	0.441	3.719	1.7%	contamination-driven
touch in box	Attacking output	0.417	3.522	1.59%	contamination-driven

Feature	Group	p50 $ z $	p99 $ z $	Frac $>$	Verdict
successful progressive passes	Passing quality	0.497	3.478	1.42%	contamination-driven
offensive duels won	Dueling	0.480	3.473	1.33%	contamination-driven
offensive duels	Dueling	0.411	3.430	1.44%	contamination-driven
sliding tackles	Defending	0.438	3.393	1.22%	contamination-driven
pressing duels	Dueling	0.490	3.317	1.2%	contamination-driven
progressive run	Progressive actions	0.442	3.280	1.22%	contamination-driven
assists	Attacking output	0.526	3.268	1.31%	contamination-driven
goals	Attacking output	0.438	3.241	1.22%	contamination-driven

Table 13: Top 20 outlier cells by $|z_{np}|$ (K=3 t-model, $\tau_{99} = 3.135$). These residuals are large relative to the model’s own tail expectations — genuine extreme seasons or persistent modelling gaps.

Rank	Player	Season	Feature	z
1	Juanmi Latasa	20/21	aerial duels	16.66
2	Álvaro Rodríguez	22/23	aerial duels	13.55
3	H. Hassan	23/24	progressive run	11.90
4	Imanol García	14/15	sliding tackles	10.84
5	Pablo Pérez	13/14	sliding tackles	10.26
6	Adrián Marín	13/14	sliding tackles	9.55
7	C. Santos	15/16	aerial duels	9.45
8	S. Weissman	20/21	successful progressive passes	-9.00
9	Juanmi Latasa	20/21	losses	8.92
10	Pablo Hervías	17/18	accelerations	8.70
11	Juanmi Latasa	21/22	aerial duels	8.61
12	Pablo Hervías	18/19	accelerations	8.61
13	Miguel Crespo	21/22	sliding tackles	8.58
14	Pablo Pérez	13/14	aerial duels	8.44
15	E. Mor	15/16	accelerations	8.44
16	M. Wakaso	14/15	sliding tackles	8.26
17	N. Kalinić	16/17	successful progressive passes	-8.04
18	Juanmi Latasa	21/22	loose ball duels	8.00
19	Edu Expósito	23/24	xg assist	7.92
20	Álvaro Giménez	18/19	successful progressive passes	-7.90

=== Old Task A vs Stage 28 comparison ===

Old tau99 (nu=2.95): 3.3709
New tau99 (nu=4.9023): 3.1345
Lighter tails (higher nu) → lower threshold → model more surprised by extremes

rate_defensive_duels_won frac > tau99: 0.0037 (old: GK-contaminated, should be lower)
Reyes 2013/14 per90_smart_passes: not found in rebuilt data (old: 14.4)
Total (n,p) cells > tau99: 1702 / 220128 (0.77%)
Expected under t(nu): ~2.65%